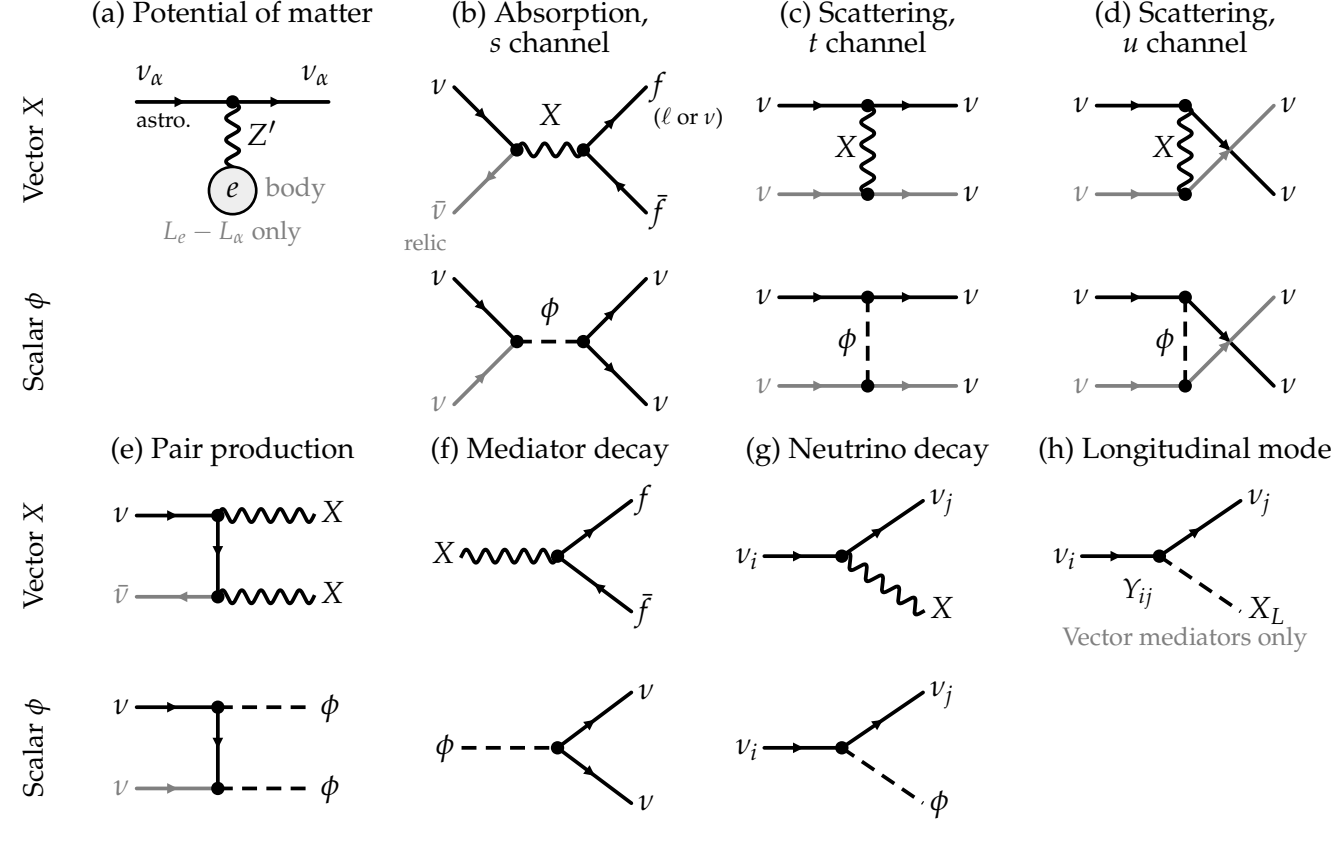

(a) Potential of matter
(b) Absorption, s channel
(c) Scattering, t channel
(d) Scattering, u channel
Vector X
Scalar φ
ν_α
ν_α
astro.
Z'
e
body
$L_e - L_\alpha$ only
ν
$\bar{\nu}$
relic
X
f
(ℓ or ν)
$\bar{f}$
ϕ
(e) Pair production
(f) Mediator decay
(g) Neutrino decay
(h) Longitudinal mode
ν_i
ν_j
Y_{ij}
X_L
Vector mediators only